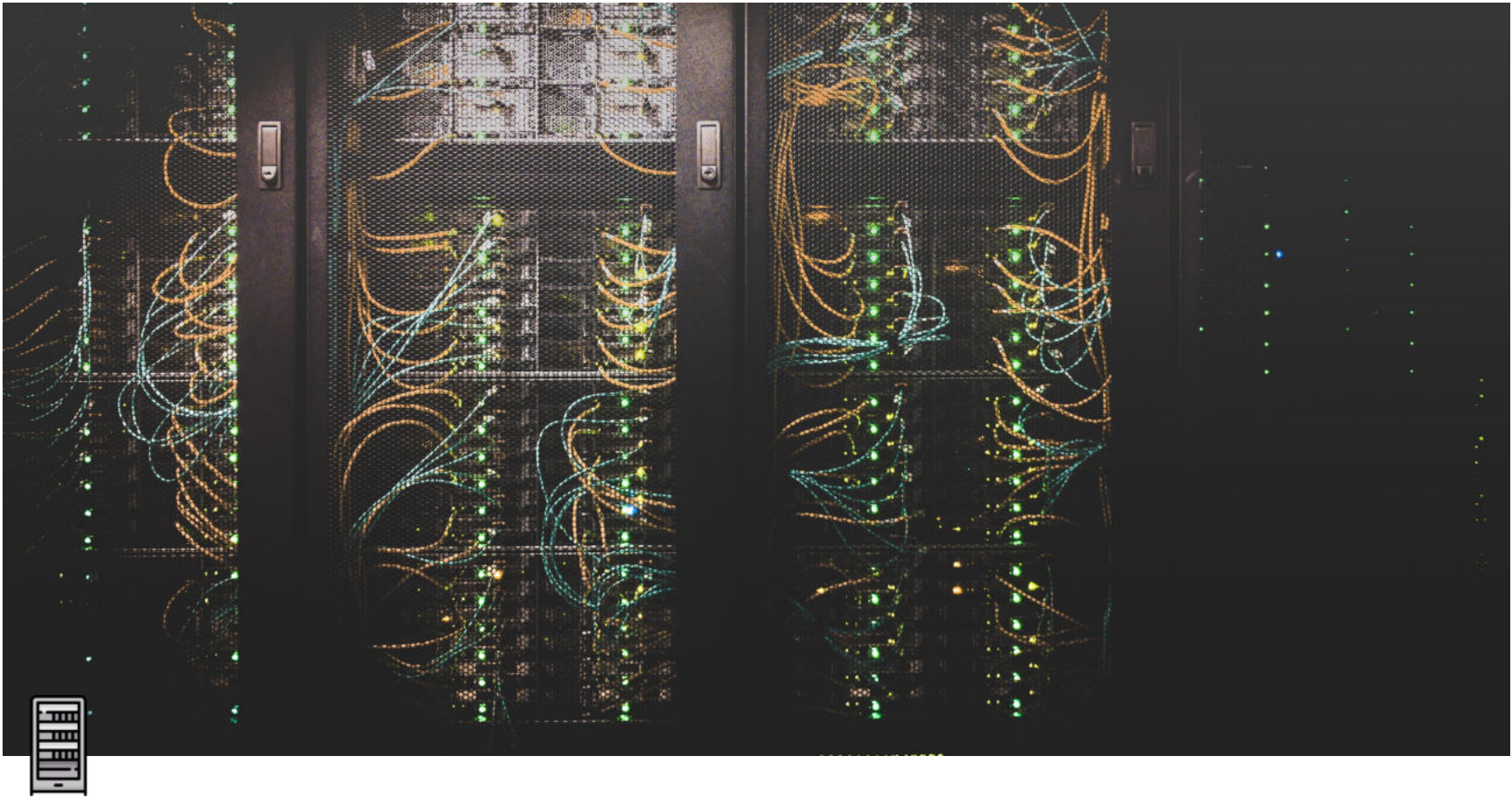




Simple Commands in Linux - 2

Type	Lecture
Date	@December 22, 2021
Lecture #	1
Lecture URL	https://youtu.be/lpe6AKk5GIk
Notion URL	https://21f1003586.notion.site/Simple-Commands-in-Linux-2-0cd2a51884c14be1a43ad24cd99c6692
Week #	2

ls

- Short and Long form of options
- Interpretation of directory as an argument
- Recursive listing
- Order of options on command line

`ls -l` will display the files & folders in the current directly, in a list

```
kashif@Zen:~$ ls -l
total 40
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Desktop
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Documents
drwxr-xr-x 3 kashif kashif 4096 Dec 21 20:16 Downloads
drwxr-xr-x 2 kashif kashif 4096 Dec 22 12:35 level1
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Music
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Pictures
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Public
drwx----- 3 kashif kashif 4096 Dec 21 20:15 snap
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Templates
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Videos
```

If we add the name of a folder after `ls -l`, it will display the files & folders of that folder

Commands to work with text files

- `less <file-name>` → To read text files, page-by-page
 - Press `q` to exit
- `cat <file-name>` → To read the entire text file, and dump them on the terminal window

- Depending on the situation, `less` can be a better way to display the contents of a file on the terminal window rather than `cat`
- `more <file-name>` → Works similarly to `less`
 - View contents of the file page-by-page
- `head <file-name>` → Displays the first 10 lines of a file
 - Takes an optional flag, `-n` followed by an integer to display that number of lines
- `tail <file-name>` → Works the opposite of `head` command
- `wc <file-name>` → Displays the number of lines (or newlines), words and bytes in a file
 - Takes optional flags like `-l` to display only the # of lines
- `which <command>` → A command to locate the commands, *heh*



- `whatis <command>` → Gives a brief description of a command
- `apropos <keyword>` → Takes in the keyword and returns a list of commands that contain the keyword in their name or in their description

```
kashif@Zen:/etc$ apropos who
w (1)          - Show who is logged on and what they are doing.
who (1)        - show who is logged on
whoami (1)     - print effective userid
```

- `help` → Displays the reserved keywords
 - Can pass a command name as optional parameter and it'll display it's help
 - Works with `shell keyword`
- `info` → Open a file (which is similar to a webpage) that lists all the commands, and we can move the cursor to any command and press **Enter** to get the manual of that command
 - Use the `<` button to go back, how do you press the `<` button? `Shift + ,`
- `type` → Know the type of a command

```
kashif@Zen:/etc$ type which
which is /usr/bin/which
kashif@Zen:/etc$ type who
who is hashed (/usr/bin/who)
kashif@Zen:/etc$ type pwd
pwd is a shell builtin
kashif@Zen:/etc$ type ls
ls is aliased to `ls --color=auto'
kashif@Zen:/etc$ type type
type is a shell builtin
```

What are aliases?

Aliases are a nickname given to a command, which may want for our convenience

How to set an alias?

```
alias <name>='<your-command>'

# Example
alias ll='ls -al'
```

How to remove an alias?

```
unalias <name>

# Example
unalias ll
```


OK, How do I know all of my aliases?

```
# Type 'alias' in the bash shell
alias
```

Multiple arguments

- Second argument
- Interpretation of last argument
- Recursion is assumed for `mv` but not for `cp`

Links

- Hard links
- Symbolic links
 - It is a bit analogous to desktop shortcuts in Windows

How to make a symbolic link?

- The command `ln` is used, with the `-s` flag

```
ln -s <source> <destination>
```

Example

```
kashif@Zen:~/Documents$ ls -al
total 20
drwxr-xr-x  4 kashif kashif 4096 Dec 30 19:27 .
drwxr-x--- 15 kashif kashif 4096 Jan  1 11:52 ..
-rw-rw-r--  1 kashif kashif   67 Dec 30 18:55 myfile.txt
drwxrwxr-x  2 kashif kashif 4096 Dec 30 19:38 python-codes
drwxrwxr-x  3 kashif kashif 4096 Dec 23 16:14 sample-proj
kashif@Zen:~/Documents$ ln -s myfile.txt shortcut_to_myfile
kashif@Zen:~/Documents$ ls -al
total 20
drwxr-xr-x  4 kashif kashif 4096 Jan  1 12:29 .
drwxr-x--- 15 kashif kashif 4096 Jan  1 11:52 ..
-rw-rw-r--  1 kashif kashif   67 Dec 30 18:55 myfile.txt
drwxrwxr-x  2 kashif kashif 4096 Dec 30 19:38 python-codes
drwxrwxr-x  3 kashif kashif 4096 Dec 23 16:14 sample-proj
lrwxrwxrwx  1 kashif kashif   10 Jan  1 12:29 shortcut_to_myfile -> myfile.txt
```

How to create a Hard link?

Just don't pass the `-s` flag to the `ln` command

```
ln <source> <destination>
```

Example

```
kashif@Zen:~/Documents$ ln myfile.txt hardlink_to_myfile
kashif@Zen:~/Documents$ ls -ali
total 24
688383 drwxr-xr-x  4 kashif kashif 4096 Jan  1 12:33 .
556357 drwxr-x--- 15 kashif kashif 4096 Jan  1 11:52 ..
655668 -rw-rw-r--  2 kashif kashif   67 Dec 30 18:55 hardlink_to_myfile
655668 -rw-rw-r--  2 kashif kashif   67 Dec 30 18:55 myfile.txt
655669 drwxrwxr-x  2 kashif kashif 4096 Dec 30 19:38 python-codes
787457 drwxrwxr-x  3 kashif kashif 4096 Dec 23 16:14 sample-proj
655696 lrwxrwxrwx  1 kashif kashif   10 Jan  1 12:29 shortcut_to_myfile -> myfile.txt
```

Notice the `inode` number for the hardlink is the same as the file

Size of files

- `ls -s` lists the files with file sizes

- `stat <file-name>` lists various other details as well as the file size
- `du <file-name>` shows the size of the file
 - Takes an optional `-h` flag to format the output in human readable form

In-memory filesystem

- `/proc`
- `/sys`

These directories are not stored on your HDD/SSD

They are located in your system memory

These contain the info about the system, so *no* `rm -rf` -ing around as `root`

A few commands

- `free` → To display the system memory stats, like used memory, free memory
 - Takes an optional `-h` flag to format the output in human readable form
- `df -h` → To know about the partitions, in human readable form